

Mustafa Fatih Can

+90 546 934 09 17

Email — LinkedIn — GitHub — Portfolio
İzmir, Turkey

Summary

Computer Engineer with 1.5 years of experience in backend development and autonomous systems. Experienced in FastAPI, ROS 2, and IoT communication (MQTT). Focused on scalable systems, geospatial data processing, and distributed architectures.

Education

Izmir Institute of Technology

2020 – 2025

B.Sc. in Computer Engineering

GPA: 3.3/4.0

Relevant Coursework: Distributed Systems, Algorithms, Robotics, Operating Systems

Experience

Computer Engineer, Dronecube

Apr 2024 – Jul 2025

- Developed backend APIs using Python (FastAPI) with event-driven architecture
- Implemented MQTT-based real-time communication
- Used Redis (caching) and RabbitMQ (async messaging)
- Processed geospatial data using GeoServer and GDAL
- Managed Docker deployments on Google Cloud Platform

Projects

Obstacle Avoidance for Agricultural Drones

- Developed modular obstacle avoidance system using ROS 2 and ArduPilot
- Implemented Rule-Based, Gap-Following, and Vector Field algorithms
- Achieved 10Hz real-time processing with <50ms latency

Distributed Database Performance Analysis

- Deployed distributed MongoDB cluster across multiple nodes
- Analyzed network delay and consistency impact

Autonomous Maze-Solving Robot

- Developed robot for solving unknown mazes
- Implemented real-time sensor-based navigation

Technical Skills

- **Languages:** Python, JavaScript, Java, SQL, C++
- **Backend & Robotics:** FastAPI, SQLAlchemy, ROS 2, ArduPilot, MAVLink, Spring Boot
- **Data & Infrastructure:** PostgreSQL, Redis, MongoDB, RabbitMQ, MQTT, Docker, GCP
- **Tools:** GeoServer, GDAL, NumPy, WebRTC, Linux/Bash, REST API, Git

Certifications

- Fundamentals of Accelerated Computing with CUDA – NVIDIA
- Social Media Content Creator – Huawei Student Developers

References

Selma Tekir

Izmir Institute of Technology

Email